



Karnataka State Police
Government of Karnataka

Powered by **I12S**

Dataathon 2026

Technology partner



Team Details

- a. **Team name:** The VK7
- b. **Team leader name:** Vikas Kumar
- c. **Team size:** 1 (solo)
- d. **Problem Statement:** Intelligent Conversational AI for KSP Crime Database

Brief about the solution

An investigation-grade identity layer, not another chatbot

WHAT IT DOES

Resolves the same human across FIR records despite transliteration, script changes, name-order flips, initials, concatenation and age-entry drift. An investigator types a name in any form — Latin or Kannada — and gets one consolidated identity with every linked record.

WHY IT MATTERS

Karnataka runs 1,100+ police stations entering names by hand. The same person becomes Ravi Patil, R. Patil, Patil Ravi, or ರವಿ ಪಾಟೀಲ್. Every missed link is a missed pattern — and a missed lead.

WHAT MAKES IT SAFE

Every match carries its reasoning, provenance and confidence. The system never fabricates a link — below threshold it flags 'verify manually' rather than guessing. In policing, a false link is wrongful suspicion on a real person.

Live query · Kannada input

ರವಿ ಪಾಟೀಲ್ → exact match finds 2 · resolved identity returns 21 across 13 districts

Opportunities

a · HOW DIFFERENT IS IT?

The obvious build for this challenge is natural language → SQL → results table. That assumes the database is already clean. It is not. We attack the layer underneath: who is actually the same person. No query sophistication recovers a record that name search cannot reach.

b · HOW DOES IT SOLVE THE PROBLEM?

Kannada → Latin transliteration, an Indian-name phonetic key (not Soundex), IDF-weighted token alignment, and average-linkage clustering that refuses to chain distinct identities through one ambiguous record. Result: 99.2% recall vs 17.9% for exact match.

c · USP

Explainability is not a feature bolted on — it is the output format. Every response ships the tables read, the method, the thresholds, the evidence per link, and what the system could not determine. We put deterministic AI where mistakes are unacceptable, and generative AI where interaction matters.

5.5×

more true links recovered vs exact match

92.2%

fewer comparisons than brute force — scales to state volume

15–30ms

warm query latency; index built once, then cached

List of features offered by the solution

Identity resolution

- Kannada → Latin transliteration, syllable-aware
- Phonetic key tuned for Indian names — not Soundex
- Name-order flips, initials, concatenation, casing & whitespace
- Age-drift tolerance via implied birth year; gender as hard signal
- IDF weighting so common surnames can't manufacture links

Investigative surface

- Consolidated case timeline per resolved identity
- Cross-district and cross-station spread of one offender
- Offence-pattern breakdown per identity
- Side-by-side contrast with exact-match results
- Per-link reasoning in plain language

Trust & governance

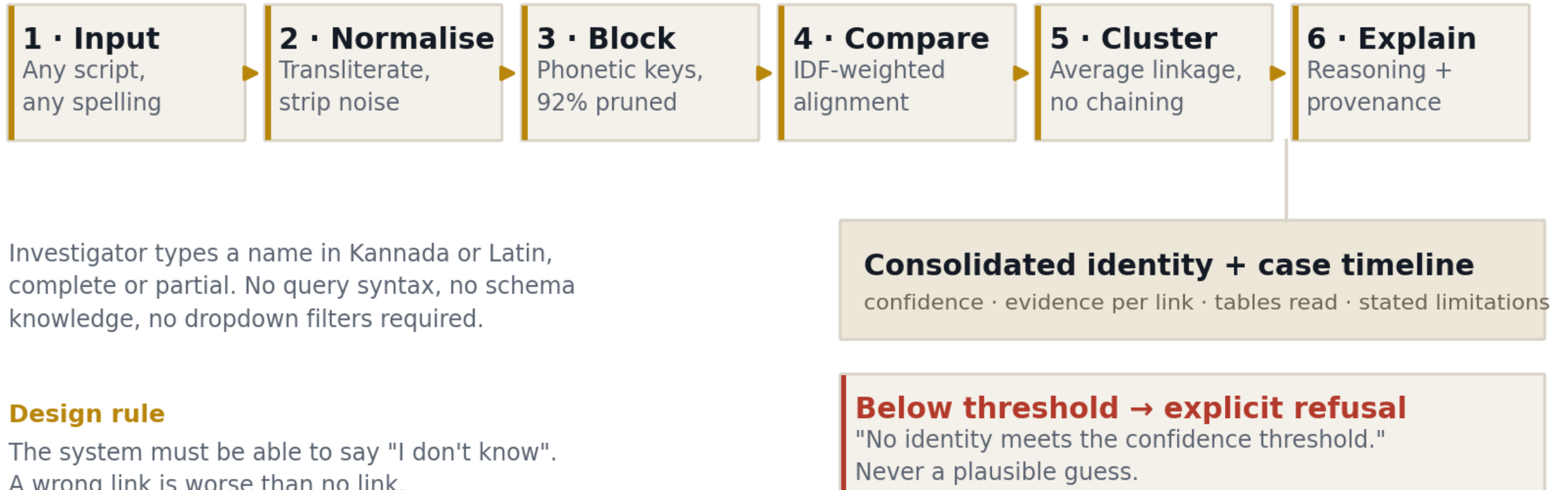
- Provenance on every response — tables, method, thresholds
- Confidence banding: high / probable / needs-review
- Never fabricates — flags 'verify manually' below threshold
- Stated limitations returned with every result

Engineering

- Phonetic blocking — near-linear, not $O(n^2)$
- Zero runtime dependencies in the engine
- Benchmarked against held-out ground truth
- Deployed end to end on Catalyst



Process flow diagram or Use-case diagram



Wireframes / Mock diagrams of the proposed solution

The interface is a case docket, not a chat window — the argument is visible at a glance.

Ravi Patil					confidence 0.97 · high-confidence
CLUSTER C0003 · 8 RECORDED SPELLINGS					
REGISTERED	NAME AS RECORDED	AGE	STATION	OFFENCE	
2023-06-05	Ravi Patil · MISSED	32	Vijayapura East PS	Vehicle Theft	
2023-04-07	R. Patil MISSED	32	Raichur West PS	Robbery	
2023-10-18	RAVI PATIL MISSED	33	Hassan West PS	Theft	
2023-07-14	R. Patil MISSED	32	Hubballi-Dharwad East PS	Vehicle Theft	
2023-01-03	RAVI PATIL MISSED	32	Shivamogga East PS	Vehicle Theft	
2024-05-31	RAVI PATIL MISSED	33	Bengaluru Rural North PS	Illegal Arms Possession	
2024-05-23	Ravi P MISSED	34	Hassan West PS	Robbery	
2024-12-29	Ravi · Patil MISSED	33	Mysuru South PS	Burglary	
2024-12-03	ರವಿ ಪಾಟೀಲ್	32	Bengaluru City East PS	Dacoity	
2024-05-18	Ravi · Patil MISSED	32	Shivamogga North PS	Theft	

WHAT A JUDGE SEES

2 vs 21

Exact match finds 2 records. The same man appears in 21, across 17 stations.

Monospace names

Trailing and double spaces render as visible dots — the invisible data-entry damage that loses a suspect.

Red = MISSED

Every row exact search would never have surfaced is flagged.

Architecture diagram of the proposed solution

PRESENTATION · CATALYST WEB CLIENT HOSTING

Investigator console

Evidence ledger · confidence banding · provenance panel · same origin as API, no CORS surface

APPLICATION · CATALYST SERVERLESS — ADVANCED I/O FUNCTION (NODE.JS)

Normalisation

Kannada translit,
token repair

Resolution engine

Blocking, IDF,
avg-linkage

Explanation layer

Reasoning, conf,
refusal

Semantic API

Governed —
no raw SQL

DATA · CURRENT

Bundled synthetic FIR corpus

1,400 FIRs · 1,663 accused · 72 stations
Schema-faithful to KSP ER diagram
Indexed once at cold start, cached

DATA · MIGRATION PATH

Catalyst Data Store · QuickML · Zia

Relational store for live FIR data,
QuickML for NL + RAG, Zia for Kannada voice.
Engine is storage-agnostic by design.

Technologies to be used in the solution

Runtime & delivery

- Node.js 24 — Catalyst Advanced I/O Function
- Express — HTTP routing inside the function
- Vanilla HTML / CSS / JS — no framework, no build
- Catalyst CLI — deployment pipeline

Resolution engine

- Custom Kannada → Latin transliteration
- Custom phonetic algorithm for Indian names
- Jaro–Winkler string similarity
- IDF term weighting
- Union-Find with average-linkage constraint
- Zero runtime dependencies — 545 lines, auditable

Data generation & evaluation

- Python 3 + SQLite — schema-faithful synthetic corpus
- Deterministic seeding — reproducible dataset
- Held-out ground-truth clusters for scoring
- Pairwise precision / recall / F1 harness

DELIBERATE NON-CHOICES

No vector DB, no embedding model, no LLM in the resolution path. Identity resolution must be deterministic and reproducible — the same query must return the same answer, and every link must be explainable to a court. LLMs are reserved for the conversational layer, where variance is acceptable.

List down Catalyst Services being used in the solution

Catalyst Serverless (Advanced I/O)	In use	Hosts the resolution engine and semantic API
Catalyst Web Client Hosting	In use	Serves the investigator console; same origin as API
Catalyst CLI	In use	Init and deployment pipeline
Catalyst Data Store	Next	Relational store for live FIR data
Catalyst QuickML	Next	Natural-language layer + RAG over case documents
Catalyst Zia Services	Roadmap	Kannada speech-to-text for field queries
Catalyst Auth + API Gateway	Roadmap	Role-based access at the data layer

Deployment is exclusively on Catalyst. Status is accurate as of submission — nothing marked 'in use' is aspirational.

Estimated implementation cost

PROTOTYPE · TODAY

~0

Runs inside the hackathon Catalyst credit allocation. One serverless function, one hosted client, no persistent store.

PILOT · ONE DISTRICT

Low

Data Store for live FIRs, function invocations, QuickML for the language layer. Index rebuild is a scheduled job, not a per-request cost.

STATEWIDE

Moderate

Dominated by Data Store volume and LLM serving. The resolution engine stays cheap — blocking keeps it near-linear.

THE ARCHITECTURAL COST DECISION

Identity resolution runs as a batch index build, not per query. Adding records re-indexes incrementally; investigator queries hit a cached structure and cost almost nothing. The alternative — an LLM call per query to reconcile names — would scale cost linearly with usage and would not be reproducible. This keeps the expensive work bounded and the frequent work free.

Snapshots of the prototype

https://ksp-crime-intel-60079832998.development.catalystserverless.in/app/index.html

RESOLVED IDENTITY — KANNADA QUERY

EXACT-MATCH SEARCH

2

records returned for "ರವಿ ಪಾಟೀಲ್". 19 further records belonging to the same person were never surfaced.

RESOLVED IDENTITY

21

records across 13 districts and 17 stations, 2023–2026.

Ravi Patil

CLUSTER C0003 · 8 RECORDED SPELLINGS

confidence 0.97 · high-confidence

REGISTERED	NAME AS RECORDED	AGE	STATION	OFFENCE
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DELIBERATE REFUSAL BELOW THRESHOLD

NAME AS YOU HAVE IT

Zzzq Nobody

Resolve identity

ರವಿ ಪಾಟೀಲ್ Ravi Patil R. Patil Khan Yusuf YusufKhan a name not on file

No identity meets the confidence threshold

No identity cluster met the confidence threshold. This is a deliberate refusal, not an empty database — narrow or correct the spelling and retry.

1,400 FIRs · 1,663 accused records

recall 99.2% vs 17.9% exact match

precision 84.7%

92.2% fewer comparisons than brute force

Prototype Performance report / Benchmarking

Measured against held-out ground truth on 1,663 accused records — scored, not asserted.

PAIRWISE ACCURACY VS GROUND TRUTH

Method	Precision	Recall	F1
Exact string match	34.1%	17.9%	23.5%
This engine	84.7%	99.2%	91.3%

Residual precision loss is data-limited, not model-limited: the corpus contains distinct people sharing a name and age. No method separates those without a corroborating identifier. 84.7% is a floor imposed by the data, not a ceiling imposed by the algorithm.

THE HONEST NOTE

A naive substring search for 'Ravi Patil' also returns Kavitha Patil — a different person — while missing R. Patil and the Kannada record. A false accusation and missed leads from one query.

99.2%

Recall — vs 17.9% exact match, 5.5× more true links

108K

comparisons instead of 1.38M — 92.2% pruned by blocking

~2s cold · 15–30ms warm

Well inside the 30-second function ceiling

Provide links to your:

1 · GitHub Public Repository

<https://github.com/reply2vikas/ksp-identity-resolution>

2 · Demo Video (3 minutes)

<https://youtu.be/mPzpEnP5KqQ>

3 · Deployed Link — Catalyst

<https://ksp-crime-intel-60079832998.development.catalystserverless.in/app/index.html>

Deployment is exclusively on the Catalyst platform, as required.

Additional Details / Future Development

1 · Semantic layer between model & schema

The LLM never writes raw SQL. It generates against governed entities and canonical joins. When the schema changes over a decade, you update one layer, not every prompt.

2 · Model-agnostic adapter

LLMs turn over every few months. Nothing in the business logic knows which model it calls. Swapping providers is a config change, not a rewrite.

3 · Graceful degradation

If the language layer is down or low-confidence, the system falls back to structured identity search and still works. No single AI dependency as a hard failure point.

4 · Audit of the queries themselves

Police search logs are themselves sensitive. Who searched whom, when, under what authority belongs in an append-only, hash-chained store from day one.

NEXT BUILD ORDER & ROADMAP

Data Store migration → QuickML natural-language layer over the resolved identity graph → criminal network graph (co-accused, shared vehicle/address, temporal & geographic trajectory) → Kannada voice via Zia → role-based access at the data layer → multi-script support (Urdu, Hindi, Tulu). Additional evidence signals — father's name, station proximity, IPC-section overlap — as weighted edges.



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THANK YOU